INTRODUCTION TO EC2 SERVICE

EC2 INSTANCE (Elastic Compute Cloud)

1. EC2 is a elastic compute cloud which is a scalable and fully flexible service which is provided by amazon AWS for launching the virtual servers, virtual machines etc.
2. it provides scalable computing capacity in AWS cloud.
3. you can use amazon EC2 to launch as many or few virtual servers as your needs, configure security and networking and manage storage.
4. it supports the autoscaling that enables you to scale up or scale down the instances.
5. it is having two storage options i.e. EBS volume and instance store. instant store is fast but it is also non-persistent storage.
6. pre-configured templates are available known as amazon machine image (AMI) to quickly launch instances with OS and software.
7. EC2 supports different instance types (General purpose, Compute optimized, Memory optimized, etc.) depending on workload.
8. By default, we create upto 20 instances per EC2 region depends on instance type (vCPU-based limits) with two default high I/0 instances.

TYPES OF EC2 INSTANCES

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1. **General Purpose (A, T, M Series)**
    1. Balanced CPU + Memory + Networking
    2. Examples: T, M, A series (latest: M7, M8, T4g)
    3. Use cases: Web apps, small DB, dev/test
2. **Compute Optimized (C Series)**
    1. High CPU performance
    2. Examples: C series (latest: C7, C8)
    3. Use cases: Gaming, batch processing, high-performance apps
3. **Memory Optimized (R, X, Z Series)**
    1. High RAM
    2. Examples: R, X, Z series (latest: R7, R8)
    3. Use cases: Big data, in-memory DB, analytics
4. **Storage Optimized (H, I, D Series)**
    1. High disk throughput & IOPS
    2. Examples: I, D, H series (latest: I7, I8)
    3. Use cases: Data warehousing, NoSQL DB

5. **Accelerated Computing (P, G Series)**
    1. Uses GPU / AI chips
    2. Examples: P, G, Inf, Trn
    3. Use cases: AI/ML, deep learning, graphics
6. **HPC Optimized (High Performance Computing)**
    1. For supercomputing workloads
    2. Examples: Hpc6a, Hpc7g
    3. Use cases: Scientific simulations, weather modelling

Status Check

Status checks are automatic checks performed by AWS to make sure your EC2 instance is running properly.

Types of status checks in AWS

1.system status

- Checks AWS infrastructure (hardware, power, network)
- Problem → AWS issue
- Fix: Stop & Start instance or wait

2.instance status

- Checks inside your instance (OS health)
- Problem → OS crash, high CPU, config issue
- Fix: Reboot or troubleshoot OS

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3.Instance reachability

- Checks whether the instance is reachable over network

If this fails:

- Security Group issue
- Network misconfiguration
- Instance stuck/not responding

Note:

1. **3/3** ----------> all ok (system / instance / reachability)
2. **2/3** ----------> One check failed (system / instance / reachability)
3. **1/3** ----------> Two checks failed (instance or infrastructure)
4. **0/3** ----------> all check failed (system / instance / reachability)

EC2 Purchasing Option

1. On Demand
2. Dedicated Instance
3. Dedicated Host
4. Spot Instance
5. Schedule Instance
6. Reserved Instance

1.On demand

1. aws on demand instances are virtual servers that runs in aws or aws relational databases service (RDS) and are purchased at a fixed rate per hour.
2. aws recommends using on-demand instances for application with short term irregular workloads that cannot be interrupted.
3. they are also suitable for use during testing and development of application on EC2.
4. with on-demand instance you only pay for EC2 instances you use.
5. the use of on-demand instances frees you from the cost and complexities of planning, purchasing, and monitoring hardware and transforms what are commonly large fixed costs into much smaller variable cost.
6. pricing is per instance-hour consumed for each instance; from the time an instance is launched until it is terminated or stopped.
7. each partial instance hour consumes will be billed per second for Linux instances and as a full hour for all other instance types.

Base point:

1. purchased at fixed rate per hour
2. only pay for consume time that we consume
3. suitable for irregular workloads and short-term workloads.
4. Commonly used for development, testing & initial deployments.
5. for Linux is bill per second and for other instances it is per hour

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2.Dedicated instance

1. Dedicated Instances run on hardware dedicated to a single AWS account.
2. They provide instance-level isolation from other AWS customers.
3. Instances are launched within a VPC environment for better control.
4. May share hardware with other instances within the same account only.
5. Suitable for workloads requiring enhanced security and compliance.

3.Dedicated host

1. Dedicated Hosts are physical servers fully dedicated to your use.
2. Provide visibility and control over physical server placement.

3. Enable usage of existing server-bound licenses (per-core, per-socket).
4. Help meet compliance and regulatory requirements.
5. Ideal for enterprises requiring hardware-level isolation and control.

4.Spot instances

1. Spot Instances utilize unused EC2 capacity at discounted pricing (up to 90%).
2. Instances can be interrupted by AWS with a 2-minute notice.
3. Suitable for fault-tolerant and flexible workloads.
4. Common use cases include batch processing, big data, and testing.
5. Interruption occurs due to capacity demand or price exceeding bid limit.

Base points

1. unused instances that we can bid on them and available at 90% of cost.
2. used for testing and deployment workloads.

Question: when would my spot instances get interrupted?

1. primary reason would be amazon EC2 capacity requirement (eg.on-demand or reserved instances)
2. secondarily if you have chosen to set a 'max spot price' and the spot price rises above this.

5.Schedule Instances

1. Scheduled Instances provide capacity reservations for recurring time windows.
2. Instances run on a defined daily, weekly, or monthly schedule.
3. You are billed for the scheduled duration, regardless of usage.
4. Ensures capacity availability during specific time periods.
5. Suitable for workloads with predictable, time-based execution patterns.

Base points

1. we can schedule for daily, weekly, and monthly basis.
2. we can reserve on advance so that it is available when we need.
3. it used for workloads that do not run continuously but do ran on regular schedule.

6.Reserved Instances

1. Reserved Instances offer discounted pricing in exchange for long-term commitment (1 or 3 years).
2. Provide significant cost savings (up to 75%) over On-Demand pricing.
3. Best suited for steady-state and predictable workloads.
4. Can include capacity reservation within a specific Availability Zone.
5. Enables cost optimization and predictable billing.

Base points

1. we can reserve for 1-3 years term

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2. it is 75% discount compared on demand.

Types of reserved instances

1. standard RI
2. convertible RI
3. schedule RI

1. **standard RI:** these provide the most significant discount (upto 75% on-demand) and are best suited for steady-state usage.
2. **convertible RI:** these provide a discount (upto 75%) and the capability to change the attributes of the RI as long as the exchange results in the creation of reserved instances of greater or equal value.
3. **schedule RI:** these are available to launch within the time window you reserve.

QUESTIONS

1. can i transfer a convertible or standard RI from one region to another?
==> No. Reserved Instances are region-specific and cannot be transferred across regions.

2. How do i change the configuration of a convertible RI?
==> You can exchange a Convertible RI for another RI with different attributes using the EC2 Management Console or AWS CLI/API, provided the new RI is of equal or higher value.

3. DO i need to pay a fee when I exchange my convertible RI?
==> No. There is no additional fee, but the price difference must be covered if the new RI has a higher value.

Launch Template

1. launch template is a way of creating multiple same ec2 instances mostly with Auto Scaling.
2. It enables you to store the ami, instance type, key pair, security groups etc.
3. It allows you to create multiple versions of the template
4. launch templates offer more advanced customization options, greater flexibility, and easier maintenance and management.

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